

Phase 2 — Rate Engine Spec

PREPARED BY Energy Hippo implementation team

Author the Northlake rate library so it fires **all 9 EEMSuite determinant math kinds** and every rate construct, lighting up the 9 Rate Analysis reports and making the Phase 1 charge lines computable from usage.

Entry path (see [entry-path-and-gaps.pdf](#)): nearly all of this is **UI-DIRECT** in the **WPF Rate Modeler** (all 9 determinant types, blocks, TOU/season/holiday schedules, ratchets, load factor, scripts, effective dates, commodity maps, multi-currency) — or bulk via **Rate Import**. RTP price point values arrive via **DATA-PATH**. **Gaps**: per-meter rate-class/**option binding** (A4) and the non-Unique "All" class (A5) have no UI — schedule #9's options can be **defined** on a rate but not **bound** to a meter through the UI.

EEMSuite model (verified): a `UtilityRate` → `UtilityRateClass` → determinants (`UtilityDeterminant` via `UtilityRateDeterminant`). **The determinant is the charge**. Each determinant has a `DeterminantTypeID` (the math kind), a `ScheduleID` (season + TOU + holiday bundle), `UtilityDeterminantBlock` rows (tiers), `UtilityDeterminantDate` rows (effective dates + RTP_PtID), optional `UtilityDeterminantRatchet` / `UtilityDeterminantScript` / `UtilityDeterminantMember`. Rates attach to meters via `UtilityMeterRates`.

The 9 determinant kinds (verified `UtilityDeterminantType`)

Type	Name	Construct
100	RTP	hourly index price × interval usage
200	Charge Period	energy/demand, tiered/blocked, TOU, seasonal
210	Charge Period Fixed Value	stepped (one rate for the whole qty by step)
300	Fixed By Day	Σ value × days
400	Fixed By Period	flat per period
500	Min Value	minimum-bill floor
600	Max Value	maximum-bill cap
700	Bill Adj	% rider/adjustment on referenced charges
800	Taxes	% tax on charges ≤ 800 (compounding)

significance (TOU): 1 All Hour, 2 Mid Peak, 3 Off Peak, 4 On Peak, 6 Super Off Peak, 7 Super On Peak.
`UtilityIntervalRule`: 1 Averaging, 2 Summing, 3 Minimum, 4 Maximum (demand uses 4 Max).

Source-model extension

implementation/synthetic-source-model.pdf rate_schedule gains determinant-level children (load-ready to Utility* tables):

- rate_determinant (rate_schedule_id, determinant_type 100–800, commodity, is_demand, interval_rule, schedule_ref, unit, use_load_factor, pre/post_script_ref)
- rate_block (rate_determinant_id, block_no, block_size/upper_bound, block_value, lf_size)
- rate_effective_date (rate_determinant_id, effective_date, block_values, rtp_point_ref, use_price_ranges, high/low_range)
- rate_schedule_calendar (schedule_ref, season {start/end month-day}, tou {hours-of-week, significance}, holiday_set)
- rate_ratchet (rate_determinant_id, current_pct, previous_pct, ratchet_months, aggregation_rule)
- rate_class_option (rate_schedule_id, class_type, option_name) — voltage/phase/tax-exempt/farm
- rate_meter_assignment (rate_schedule_id, meter_id, rate_class_option_id, bill_scenario, start_date)

The 10 Northlake rate schedules

Designed so that, between them, every UtilityRates_Run_NNN proc and every construct fires. "Extends" = enrich an existing provider rate; "new" = a new rate object/option in the library.

1. NL-RES-Tiered — Cedar Row residential electric (extends VED for Cedar Row)

Determinants: **400** customer charge; **200** inclining-block energy (≥ 3 blocks: 0–500 / 500–1500 / 1500+ kWh); **500** minimum bill; **800** state + municipal tax (two members). Fires 200(blocks), 400, 500, 800; tier clamping.

2. NL-GS-TOU-Seasonal — campus commercial electric (flagship) (extends VED-TOU-GS)

Determinants: **400** facility charge; **200** On-Peak energy (sig 4) + **200** Off-Peak energy (sig 3) on TOU schedules; **200** summer demand + **200** winter demand (IsDemand=1, IntervalRule 4) on seasonal schedules; a **holiday set** mapping weekday holidays to off-peak; **700** % public-purpose rider; **200** per-unit renewable rider; **800** utility-users tax. Fires TOU, seasons, holiday calendar, both rider styles, demand, the full significance taxonomy. This is the rate the Phase 1 VED charge lines decompose against.

3. NL-LG-Ratchet — Science Center large demand (new VED large-GS option)

Determinants: **200** demand with a rate_ratchet (80% / 11-month look-back); **200** energy with **load factor** (use_load_factor, lf_size); **400** fixed. Needs ≥ 12 months interval with a strong summer peak so the winter ratchet floor binds. Fires the ratchet path (almost unseeded product-wide).

4. NL-CONTRACT-2Part — base/excess contract demand (new, Central Plant Electric Service)

Determinants: **200** base contract demand + **200** excess demand (split by a **pre/post** rate_determinant.script); base + excess energy; **400** fixed; **800** tax. Fires determinant scripts + two-part contract demand.

5. NL-RTP-Index — real-time pricing (new VED RTP option, Student Center)

Determinants: **100** RTP referencing an **hourly price-signal point** (`rtp_point_ref` → a new `reference_channel` carrying hourly \$/kWh); a second **100** with `use_price_ranges=1` + high/low band; **400** delivery charge. **The only way to fire** `UtilityRates_Run_100`. Requires a new synthetic hourly price channel (add to `data/scenarios/demo-university-v1.yaml` `referenceChannels`).

6. NL-STEP-Demand — stepped + capped (new, Town Center anchor — see Phase 4)

Determinants: **210** stepped demand (one rate by which step the kW lands in); **600** maximum-bill cap on the subtotal. **The only way to fire 210 and 600** (both zero-seeded product-wide).

7. NL-GAS-FuelAdj — gas with monthly fuel adjustment (extends SGU-GN-1)

Determinants: **300** fixed-by-day customer charge; **200** seasonal commodity (winter/summer) + **200** transport with **unit conversion** (`ccf` → therm, factor 1.027); a **"Fuel Cost Adjustment" 200** with **monthly** `rate_effective_date` **rows** (a new `BlockValue` per month → effective-dated proration within a bill period); **800** tax. Fires 300, unit conversion, effective-dated proration.

8. NL-MULTI-Commodity — water/sewer/CHW/steam (extends RCU + NTP)

Determinants: water tiered (**200** blocks), sewer as **700** % of water, chilled-water demand in **tons** (**200** `IsDemand`), steam in **klb** (**200**). Uses `UtilityCommodityMap` for ENERGY STAR component mapping (`rp_RateComponentMap`). Fires multiple `MeasureTypes` + unusual units (tons, klb).

9. NL-OPTIONS-Phase — rate-class options (new VED variant)

An "All" base class + selectable option classes: **Primary vs Secondary voltage, Single/Three-phase, Tax-Exempt, Farm Discount** (`rate_class_option`). Meters pick options via `rate_meter_assignment.rate_class_option_id`. Fires class assignment + `rp_RateClassAssignment`.

10. NL-MULTI-Currency — cross-currency (optional)

A rate with `CurrencyID` = C\$ billed against a US\$ reporting unit profile. Fires currency conversion. (Optional completeness; flag as synthetic.)

Meter assignment

Each schedule attaches via `rate_meter_assignment` over ≥13 months:

- NL-GS-TOU-Seasonal → Admin Hall, Library electric (monthly TOU registers) + via interval for Science/Student is NL-LG-Ratchet/NL-RTP-Index.
- NL-RES-Tiered → the Cedar Row A and Cedar Row B Electric Master meters.
- NL-LG-Ratchet → Science Center electric. NL-RTP-Index → Student Center.
- NL-CONTRACT-2Part → Central Plant Electric Service (Northlake Thermal Plant company). NL-STEP-Demand → Town Center anchor.
- NL-GAS-FuelAdj → all SGU meters. NL-MULTI-Commodity → RCU + NTP meters.
- NL-OPTIONS-Phase → an Admin Hall variant (Primary voltage, Tax-Exempt option).

Unsupported constructs (synthetic stand-ins — document explicitly)

The engine cannot model these natively; represent and label as stand-ins:

- **Power factor** → precompute the adjustment quantity into a billing channel or a post-script (the VED `power_factor_adjustment` line).
- **Coincident peak / 4-CP** → script the determinant value or mark CP hours via an RTP-style point.
- **CPP / critical-peak events** → a TOU schedule hard-coding event hours, or RTP price spikes on event days.
- **Net-metering banking** → no cross-month credit balance; approximate with a per-month negative credit block (HOS `net_metering_credit`).

expected Assertions

- `every_run_proc_fires` — each `UtilityRates_Run_100..800` executes $\geq$ once across the library.
- `rtp_priced_from_point` — NL-RTP-Index bill = $\sum(\text{interval kWh} \times \text{hourly price})$; the price-range variant caps at high/low.
- `stepped_and_cap` — NL-STEP-Demand picks one step rate; 600 caps the subtotal when it would exceed the cap.
- `ratchet_floor_binds` — a winter NL-LG-Ratchet billed demand = 80% of the prior summer peak when current demand is lower.
- `tiers_and_min_bill` — NL-RES-Tiered blocks clamp to usage; a low-usage month tops up to the 500 minimum.
- `tax_compounds_in_order` — two 800 taxes apply in determinant order over members.
- `rate_comparison_shadow_bills` — `rp_RateComparison` recomputes a meter on an alternate schedule and diffs vs the baseline scenario.
- `class_option_assignment` — `rp_RateClassAssignment` shows the Primary/Tax-Exempt options on the NL-OPTIONS-Phase meter.